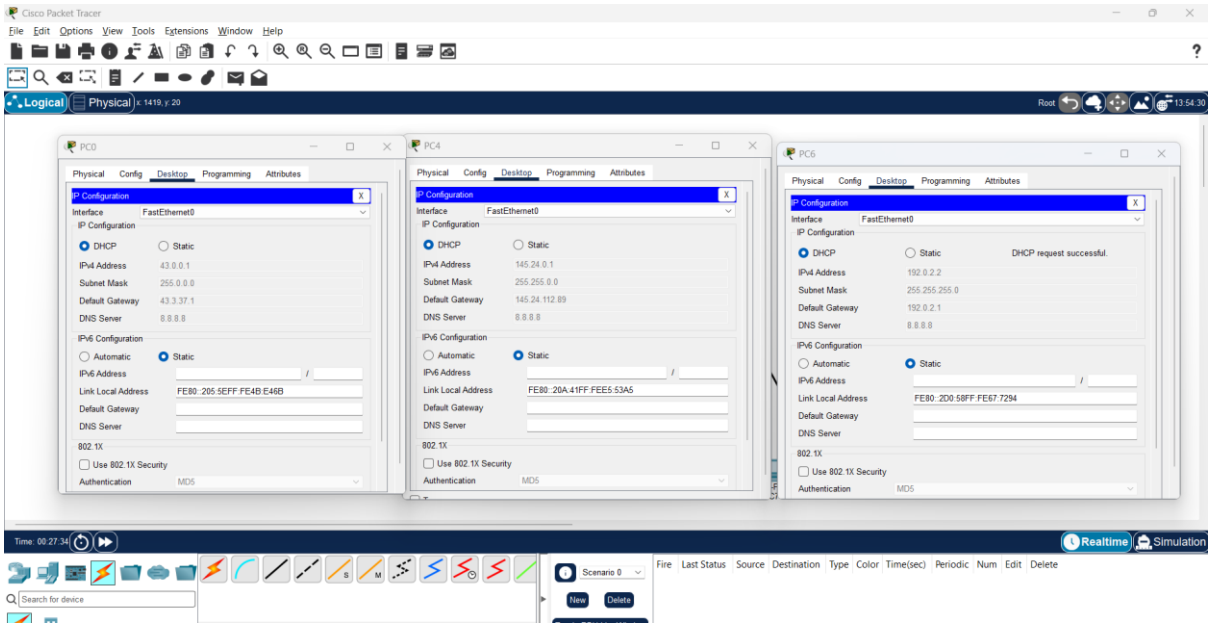
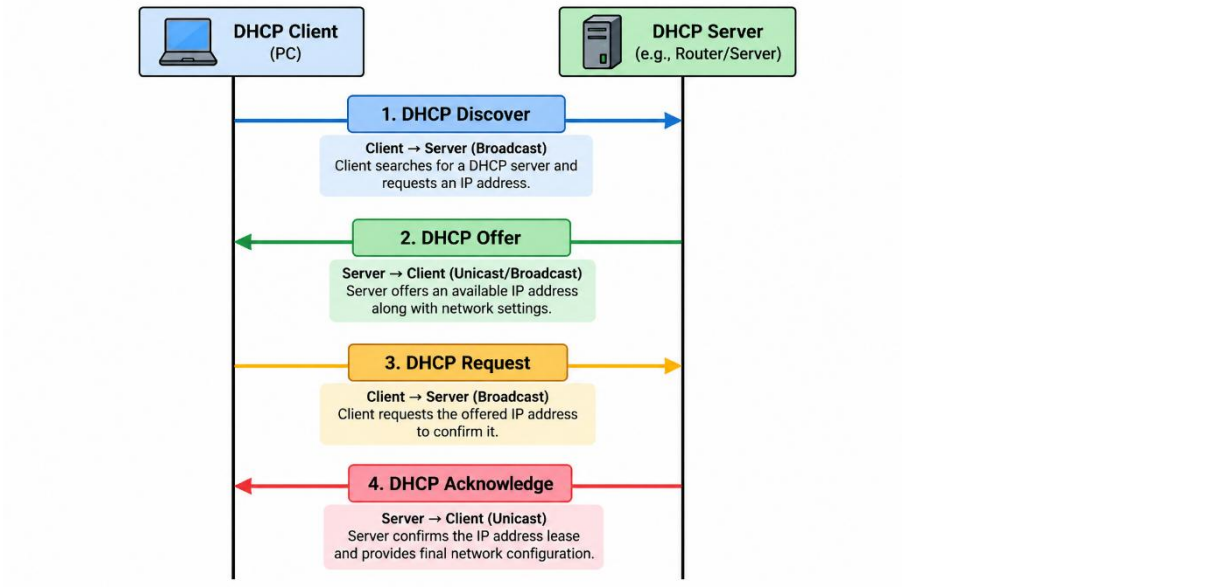
Day – 2

1.In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.



In Cisco Packet Tracer, the client PCs were configured to obtain IP addresses automatically using DHCP. The PCs received IP addresses from the DHCP pool, allowing them to communicate over the network without manual IP configuration.

2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.



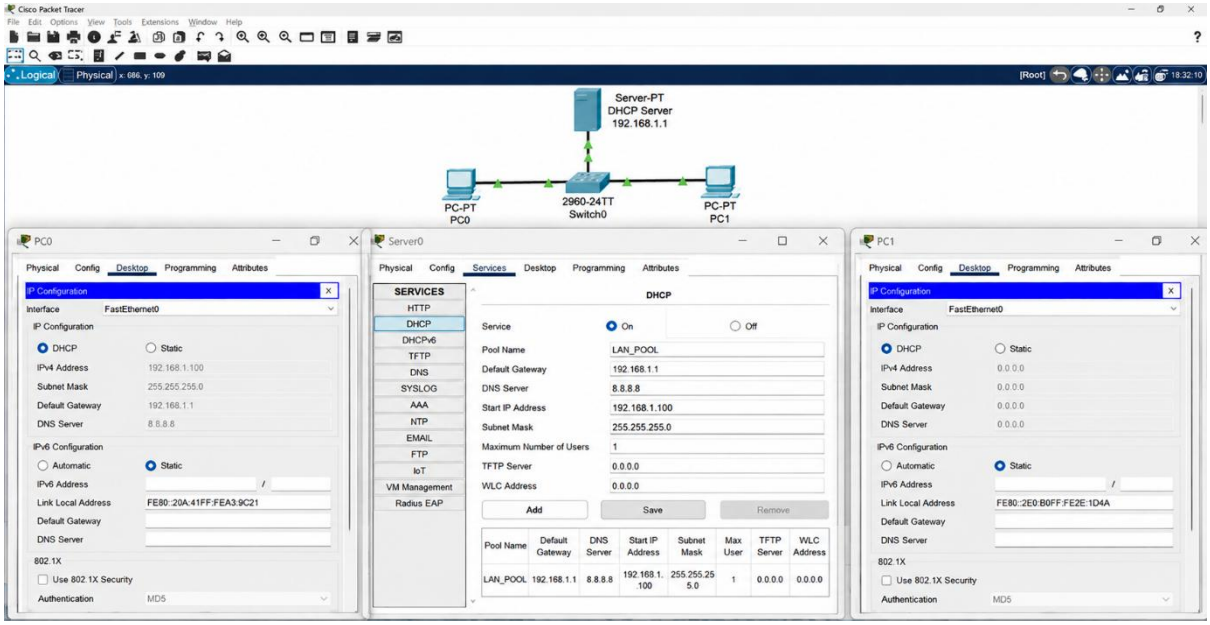
Discover: Client → Server — The client searches for a DHCP server.

Offer: Server → Client — The server offers an available IP address.

Request: Client → Server — The client requests the offered IP address.

Acknowledge: Server → Client — The server confirms the IP address lease and provides the network configuration.

3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.



The DHCP server is configured with only one available IP address (192.168.1.100). PC0 receives the available IP address, while PC1 cannot obtain an IP address because the DHCP pool is exhausted.

4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

DHCP Configuration

Configure the server in Cisco Packet Tracer using Services → DHCP.

Setting	**Value**

Pool Name	RESTAURANT_WIFI
DHCP Service	On
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8
Maximum Users	50

Connect the server and two client devices to a switch. On each client, select Desktop → IP Configuration → DHCP. Verify the assigned IP addresses.

Assigned IP Addresses

Device	Assigned IP Address	Method
Laptop	192.168.1.10	DHCP
Smartphone (generic PC)	192.168.1.11	DHCP

These are example addresses; record the actual addresses shown in your Packet Tracer simulation.

Why DHCP Is Useful

DHCP automatically assigns IP addresses to customer devices connecting to a restaurant's Wi-Fi, eliminating the need for manual configuration. It reduces IP address conflicts and makes it easier to manage many devices as customers arrive and leave.